

Assignment 3: Bayesian Networks with bnlearn

R version (bnlearn for R)

Overview

In this assignment you will use the R package **bnlearn** to analyze survey data on preferences among three long-distance telephone providers. You will build and compare five Bayesian network classifiers, including naive Bayes, and explain in your own words what each package function does and how it maps to the theory in Lecture Set 4 (Bayes networks, slides 14–44).

By the end you should be able to:

1. Translate a DAG into a factorization of the joint distribution (slides 15–20).
2. Fit a naive Bayes classifier with Dirichlet priors, compute posterior classification probabilities by hand, and verify them in software (slides 23–24, 31–38).
3. Explain how the software's *equivalent (imaginary) sample size* corresponds to the global precision α and the “initial ignorance” prior $\alpha_{ijk} = \alpha / (C_i Q_i)$ (slide 33).
4. Compute log marginal likelihoods and use them to compare DAGs (slides 39–40).
5. Compare competing classifiers by their posterior classification probabilities and log marginal likelihoods, and judge what a violated naive Bayes assumption does to classification.

In this Assignment you will work as a group. Each group submits one write-up and one script, and presents its work in class (guidelines to follow). Another group is doing the same assignment in Python; in class the two groups will compare results.

Software

Install the package from CRAN with `install.packages("bnlearn")` and report the version you used (`packageVersion("bnlearn")`). **Documentation:** `help(package = "bnlearn")` and bnlearn.com. You also need the package `gRain` (`install.packages("gRain")`), which `bnlearn` uses for exact inference with `predict(..., method = "exact")`. The base `plot()` method is enough for drawing networks.

The data

The file `carrier_data.csv` holds 877 survey responses. Respondents evaluated their current long-distance provider and at least one alternative on several attributes. The providers are anonymized as A, B and C.

Variable	Meaning	Coding in the file	Recode as
Y	Preferred long-distance provider	1 = A, 2 = B, 3 = C	3 categories
X1	Satisfaction with the provider's reputation as an industry leader	0 = low, 1 = high	low / high
X2	Satisfaction with price	0 = low, 1 = high	low / high
X3	Satisfaction with variety of services	0 = low, 1 = high	low / high

Recode X1–X3 as factors with levels `low` and `high`, and Y as a factor with levels A, B, C. `bnlearn` requires every discrete variable to be a `factor`; explain in your write-up why.

Using generative AI

You may use a generative AI tool (ChatGPT, Claude, Copilot, Gemini, etc.) to learn the package and to help write code. You are responsible for every line you submit. Please be sure that you check every function and argument against the package help (`?hc`, `?bn.fit`, `?score`, ...).

Part A — A guide to the functions you use, tied to the lecture

Write a short guide to the package functions you use in Part B, organized by the lecture topic they implement. For each function give: (a) what it does, in one or two sentences of your own; (b) the arguments you used and what they mean; (c) the lecture concept and slide it corresponds to, with notation matched (which quantity in the software is N_{ijk} , θ_{ijk} , α or $p(D \mid M)$); and (d) a one-line example from your own code. The table lists the functions you will need and the questions your guide must answer. Keep the guide to about two to three pages.

Lecture topic (slides)	Functions	Questions your guide must answer
DAGs and factorization (15–19, 23)	<code>model2network</code> , <code>modelstring</code> , <code>naive.bayes</code> , <code>nparams</code>	How does a model string encode $p(\mathbf{y}) = \prod_i p(y_i \mid \Pi_i)$? Give the model strings for M1–M5. How does the network returned by <code>naive.bayes()</code> relate to the model string of M2?
Parameter learning (29–33)	<code>bn.fit</code> with <code>method = "mle"</code> and <code>"bayes"</code> , argument <code>iss</code> ; <code>coef</code>	Show that <code>"mle"</code> gives N_{ijk}/N_{ij} (slide 30). Show that <code>"bayes"</code> with <code>iss = α</code> returns the Dirichlet posterior mean with $\alpha_{ijk} = \alpha / (C_i Q_i)$ (slides 32–33).
Classification probabilities (24, 38)	<code>predict</code> with arguments <code>prob</code> and <code>method("parents", "bayes-lw", "exact")</code> ; <code>cpquery</code>	Where is $p(y \mid x_1, x_2, x_3, D)$ of slide 24 computed? What do the <code>predict()</code> methods <code>"parents"</code> , <code>"bayes-lw"</code> and <code>"exact"</code> do, and which is appropriate when Y has no parents? Why do <code>cpquery()</code> and <code>"bayes-lw"</code> give slightly different answers on each run?
Marginal likelihood (40)	<code>score</code> with <code>type = "bde"</code> and <code>"bic"</code> ; arguments <code>iss</code> , <code>by.node</code>	Which score equals $\log p(D \mid M)$ on slide 40? Why is it a sum over nodes?

Part B — Analysis of the carrier data

Use global precision $\alpha = 12$ throughout unless a task says otherwise. Set a random seed wherever results are simulated.

B1. Exploration

Read and recode the data. Report the four-way contingency table and the cross-tabulations of Y with each X. Which attribute appears most related to the preferred provider?

B2. Naive Bayes by hand, then in software

1. Draw the naive Bayes DAG with Y as the class node. Write its factorization and list its parameter vectors $\theta_1, \theta_{2j}, \theta_{3j}, \theta_{4j}$ in the notation of slide 26.
2. Give the Dirichlet prior parameters implied by $\alpha = 12$ and the posterior Dirichlet for each parameter vector, in the format of slide 36.
3. Compute by hand the posterior classification probabilities for a customer with high reputation, low price and high services satisfaction (format of slide 38).
4. Reproduce steps 2 and 3 with `naive.bayes()`, `bn.fit()` and `predict(..., prob = TRUE)`, and again with `cpquery()`. Explain why `cpquery()` gives a slightly different number each time.

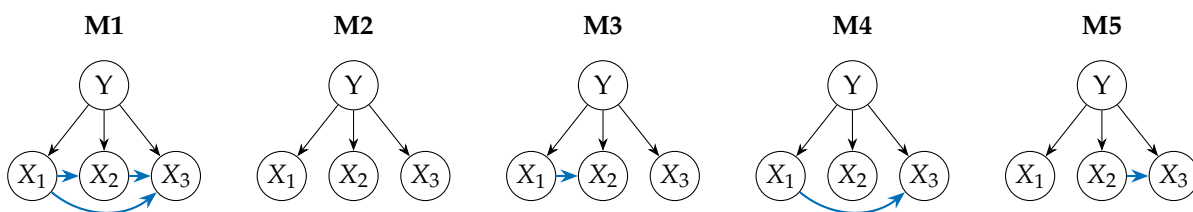
B3. Prior sensitivity

Refit the naive Bayes classifier with $\alpha = 1, 12, 100$ and 1000 and report the classification probabilities for the customer in B2.3. Explain what happens as α grows, using the form of the posterior mean.

B4. Five candidate classifiers

Consider the following five models for predicting Y . In every model Y is a parent of X_1 , X_2 and X_3 ; the models differ in which attributes remain dependent *given* Y .

Model	Conditional dependence structure given Y
M1	Full dependence: X_1, X_2 and X_3 are all dependent on each other (and on Y).
M2	Naive Bayes: X_1, X_2 and X_3 are mutually independent given Y .
M3	X_3 is independent of (X_1, X_2) given Y ; X_1 and X_2 are dependent given Y .
M4	X_2 is independent of (X_1, X_3) given Y ; X_1 and X_3 are dependent given Y .
M5	X_1 is independent of (X_2, X_3) given Y ; X_2 and X_3 are dependent given Y .



Work through the following three tasks for each of the five models.

1. **Factorization.** Write down the full joint distribution $p(y, x_1, x_2, x_3)$ as a product of the conditional distributions implied by the DAG, as on slides 16–19. For each factor, list the parent configurations.
2. **Estimation and posterior classification probabilities.**
 - (a) Build each DAG with `model2network()` and estimate its parameters with `bn.fit(method = "bayes", iss = 12)`. Report the Dirichlet posterior for each parameter vector (format of slide 36) for at least one model other than naive Bayes.

- (b) Obtain the posterior classification probabilities $p(Y \mid x_1, x_2, x_3, D)$ for all 8 attribute profiles under each model, and the predicted class. Use `predict(..., prob = TRUE)`, and compare its three prediction methods: the default `method = "parents"`, `method = "bayes-lw"` and `method = "exact"`. Explain what each method does, why the default is not appropriate here (in all five DAGs Y has no parents), and why "bayes-lw" gives slightly different answers on each run (set a seed). Use `method = "exact"` for the rest of B4.
- (c) Compare the five models: put their classification probabilities side by side, and report each model's confusion matrix and accuracy. Where do the models disagree, and why? Are there profiles where the top two classes are tied or nearly tied?
3. **Comparison by log marginal likelihood.** Compute the log marginal likelihood $\log p(D \mid M_g)$ of each model at $\alpha = 12$ (slide 40) with `score(..., type = "bde", iss = 12)`, and verify one node's contribution by hand. Convert the log marginal likelihoods to posterior model probabilities under equal prior probabilities $p(M_g) = 1/5$. Which model is best supported by the data, and how strongly? Does the model with the highest marginal likelihood also classify best?

Interpretation

In at most one page, state what the analysis says about which attributes drive provider preference, and what that implies for providers A and C.

Submit before class (or post to BB) on October 8, 2026: (1) a PDF write-up containing Part A and the answers to Part B with tables and figures; (2) one commented R script that runs from top to bottom in a fresh session and reproduces every number in the write-up.

References

- Lecture Set 4, DNSC 8328, slides 14–44.
- Scutari, M. (2010). Learning Bayesian networks with the bnlearn R package. *Journal of Statistical Software*, 35(3), 1–22.
- Package documentation and examples: bnlearn.com.